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1. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

In response to an increase in the frequency of thunderstorms that produce damaging hail, some apple growers have started to cover their apple orchards with large nets. While these nets protect the apple trees from hail damage, the nets also decrease the amount of light that reaches the trees. In an attempt to reduce the impact of shading by the nets, researchers investigated the ability of different ground covers placed around the base of the trees to reflect light back to the apple trees. An apple orchard with genetically identical apple trees was divided in half. A reflective ground cover was applied to one half of the orchard, while grass was left as the ground cover for the other half. A sample of apples was taken from every tree. The percent of light reflected by the two ground covers was measured at three distances above the ground (Figure 1).

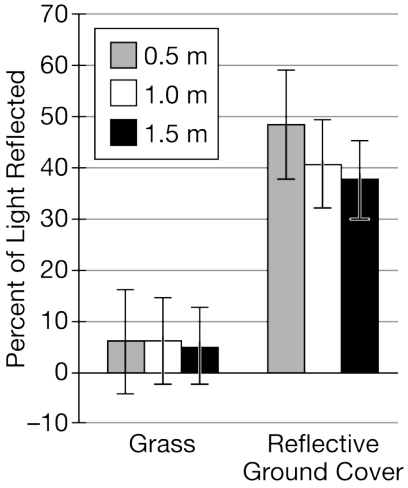


Figure 1. Percent of light reflected by ground covers, measured at three distances above the ground. Error bars equal $\pm 2\mathrm{SE}_{\bar{x}}$.

The amount of light that apple trees receive changes the expression of a certain transcription factor. This transcription factor affects the transcription rate of the gene encoding the enzyme phenylalanine ammonium-lyase (PAL). The PAL enzyme is required for the production of two types of pigments, anthocyanins and flavonoids, which contribute to the coloration of apples. Light also affects the activity of the PAL enzyme by a signal transduction pathway (Figure 2).

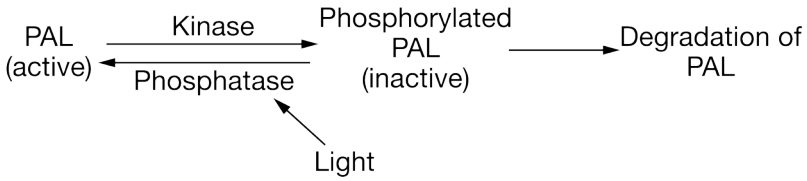


Figure 2. Simplified model of the signal transduction pathway leading to PAL activation and breakdown

The researchers measured the average concentrations of anthocyanins and flavonoids in apples from trees grown with both types of ground cover (Table 1).

TABLE 1. CONCENTRATION OF ANTHOCYANINS AND FLAVONOIDS IN APPLES

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Type of Pigment	Concentration of Pigment (mean $\pm 2\mathrm{SE}_{\bar{x}}$ in $\mathrm{nmol\ cm^{-2}}$) in Apples Grown with Grass Ground Cover	Concentration of Pigment (mean $\pm 2\mathrm{SE}_{\bar{x}}$ in $\mathrm{nmol\ cm^{-2}}$) in Apples Grown with Reflective Ground Cover
Anthocyanins	14.51 ± 1.9	24.10 ± 3.0
Flavonoids	29.24 ± 3.6	44.50 ± 5.7

(a) Using the information provided in Figure 2, **explain** the role of light in the activation of the PAL enzyme.

(b) **Identify** the variable in the researchers' experiment that is dependent on the percent of reflected light the apple trees receive. **Justify** the use of genetically identical apple trees as a controlled factor in the experiment. Using the data provided, **describe** the effect of the different ground covers on the percent of light reflected. **Calculate** the percent change in the average concentration of flavonoids in apples grown with the reflective ground cover compared with the apples grown with the grass ground cover.

(c) Researchers also study apple trees that have been genetically modified so that the DNA binding domain of the transcription factor that stimulates expression of the PAL enzyme has been deleted. The researchers repeat the ground cover experiment with the genetically modified apple trees. **Predict** the effect of the two ground covers on pigment concentrations in the genetically modified apples. Provide reasoning to **justify** your prediction.

(d) Based on the information provided in Figure 2, **predict** the effect of a constantly active kinase on flavonoid pigment production. In leaves, the PAL enzyme is also required for the production of other pigment molecules that are capable of absorbing UV and other radiation from the sun. **Explain** how the relationship between light and pigment production might serve as a protective mechanism for the cells of the apple tree. **Explain** how the PAL enzyme interacts with its substrate in the absence of light.

2. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

In plants, a critical step in the pathway leading to the synthesis of the amino acid tryptophan includes the synthesis of an intermediate molecule, 5-enolpyruvylshikimic acid-3-phosphate (EPSP). The reactants in this step are shikimic acid-3-phosphate ($\mathrm{S3P}$) and phosphoenolpyruvate (PEP), and the reaction is catalyzed by the enzyme EPSP synthase (Figure 1). Compound X, a chemical used to kill weeds, can affect EPSP synthesis by interacting with EPSP synthase.

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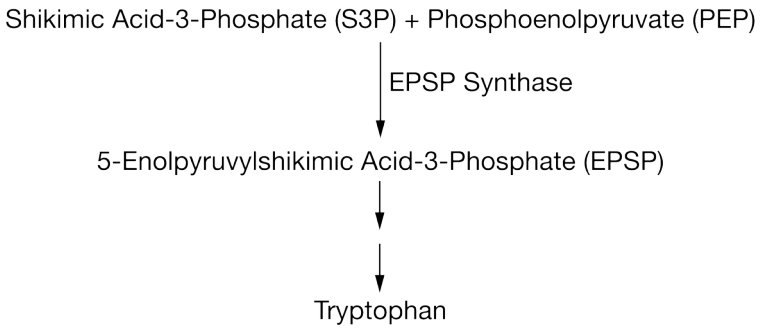


Figure 1. Reaction between shikimic acid-3-phosphate (S3P) and phosphoenolpyruvate (PEP), catalyzed by the enzyme EPSP synthase. Compound X can affect EPSP synthesis by interacting with EPSP synthase.

Scientists investigated the effect that Compound X has on EPSP synthase activity by performing an experiment to examine the relationship between EPSP synthase activity, PEP concentration, and Compound X concentration (Figure 2).

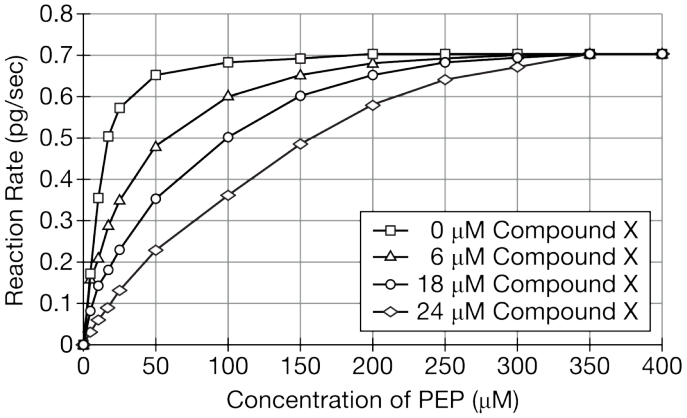


Figure 2. Rate of the reaction catalyzed by a specific amount of EPSP synthase with a specific amount of S3P, increasing concentrations of PEP, and four concentrations of Compound X

Some populations of the plant *Lolium rigidum* have evolved resistance to Compound X. To investigate the inheritance of this resistance, researchers crossed true-breeding plants from a Compound X-susceptible (S) population with true-breeding plants from a Compound X-resistant (R) population. In half of these crosses, pollen from S plants was used to fertilize ovules in R plants. In the other half, pollen from R plants was used to fertilize ovules in S plants. Survival of parental and F_1 seedlings, differentiated by the maternal phenotype, in response to varying concentrations of Compound X is shown in Figure 3.

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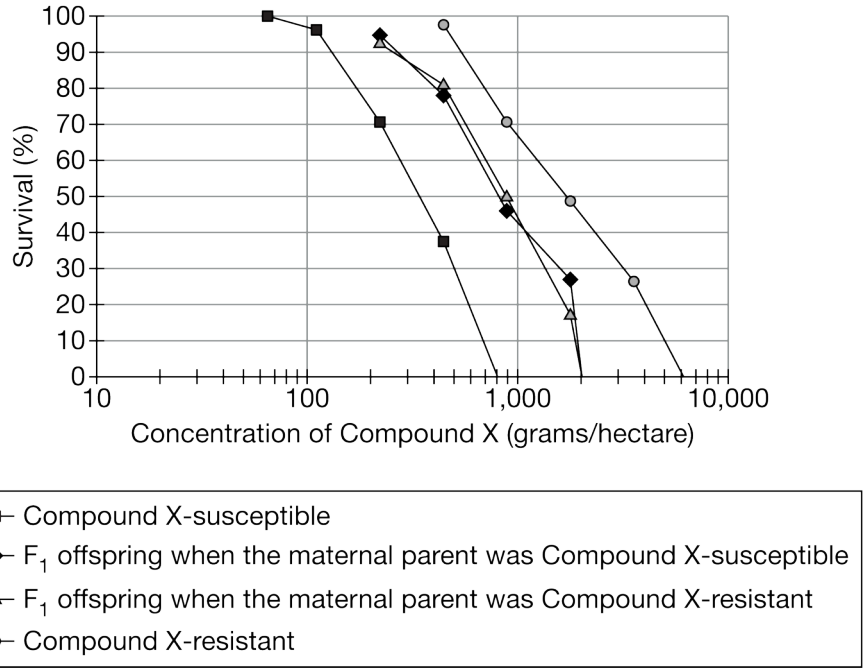


Figure 3. Percent survival of four populations of *L. rigidum* exposed to various concentrations of Compound X shown in a log scale. Data are an average from one experiment with four replicates conducted on seedlings growing in pots.
$1\ \mathrm{Hectare} = 10{,}000\ \mathrm{m}^2$

(a) Compound X competes with PEP for binding to the active site of EPSP synthase. **Explain** how increasing the concentration of PEP can reverse the inhibitory effect of Compound X on the reaction catalyzed by EPSP synthase.

(b) **Identify** a control group in the experiment shown in Figure 2. **Identify** the dependent variable in the experiment shown in Figure 3. Using the data provided in Figure 3, **identify** the concentration of Compound X, in $\mathrm{grams/hectare}$ required to kill 60% of F_1 offspring when the maternal parent was Compound X-susceptible. Using the graph in Figure 2, **calculate** the mass, in pg, of EPSP that would be produced by this reaction in one minute at a PEP concentration of $200\ \mu\mathrm{M}$ and a Compound X concentration of $0\ \mu\mathrm{M}$.

(c) Researchers claim that a single gene is responsible for the Compound X-resistance phenotype. In a follow-up experiment with *L. rigidum*, the researchers crossed an F_1 female with a male from the original susceptible (S) population. The resulting offspring were treated with $300\ \mathrm{grams/hectare}$ of Compound X. Based on the data presented in Figure 3, **predict** ONE genotype of the resulting offspring and the percent survival of the offspring with this genotype after treatment with Compound X. A student claims that the allele for the Compound X-resistance gene exhibits incomplete dominance over the Compound X -susceptible allele. Based on data provided in Figure 3, provide reasoning to **support** this claim. Researchers discovered a variant of the Compound X-resistance gene in *L. rigidum* in which there was a two base pair deletion early in the coding region of both copies of the gene. **Predict** the most likely effect on the phenotype of the plant.

(d) Researchers have engineered Compound X-resistant corn by inserting a Compound X-resistance gene from a prokaryote into the corn genome. **Explain** how scientists could ensure that the gene would be reliably transmitted to the offspring by either the pollen or the ovule. **Explain** why the $EPSP$ synthase gene and the pathway it regulates can exist in both prokaryotes and corn plants.

3. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

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Saccharomyces cerevisiae, also known as baker's yeast, is a single-celled eukaryote that can make ATP both aerobically and anaerobically. To obtain a readily accessible form of chemical energy for cellular activities, many strains of yeast convert glucose into ethyl alcohol and carbon dioxide.

Researchers conducted an experiment to determine the effect of temperature on the rate of glucose conversion to ethyl alcohol and carbon dioxide. They added an equal mass of yeast cells to equal volumes of a glucose solution in a series of sealed test tubes. Groups of tubes were incubated at temperatures ranging from $28^{\circ}C$ to $40^{\circ}C$, the optimal range for yeast growth. The researchers measured carbon dioxide production in the tubes.

(a) **Describe** the importance of fermentation to eukaryotic cells such as yeast.

(b) **Identify** the independent variable in the experiment.

(c) **Predict** the relative rates of glucose breakdown at $4^{\circ}C$ AND at $56^{\circ}C$ compared with the rate at $37^{\circ}C$.

(d) **Provide reasoning** to support your prediction in part (c).

4. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

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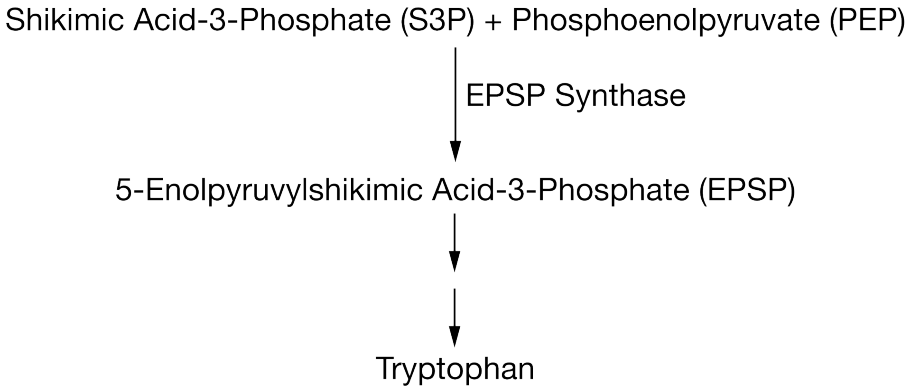


Figure 1. Reaction between shikimic acid-3-phosphate (S3P) and phosphoenolpyruvate (PEP), catalyzed by the enzyme EPSP synthase. Compound X can affect EPSP synthesis by interacting with EPSP synthase.

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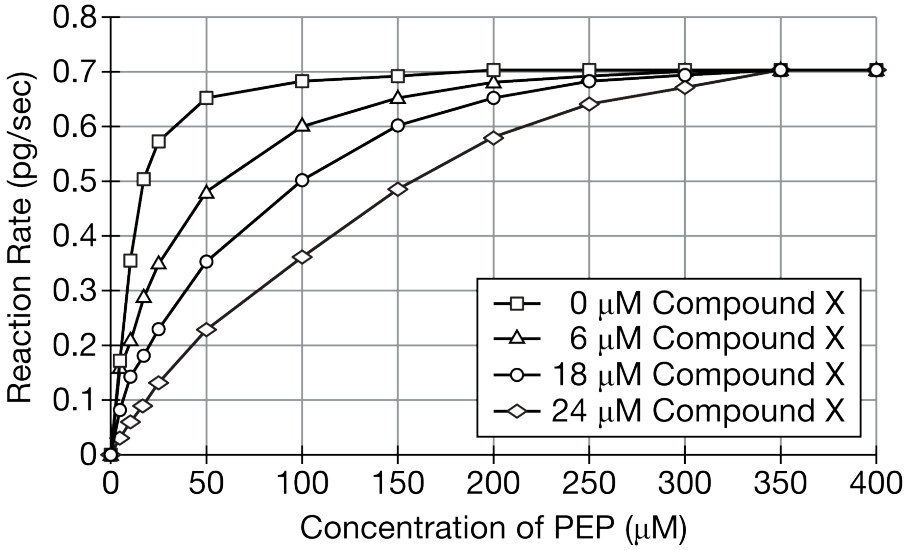


Figure 2. Rate of the reaction catalyzed by a specific amount of EPSP synthase with a specific amount of S3P, increasing concentrations of PEP, and four concentrations of Compound X

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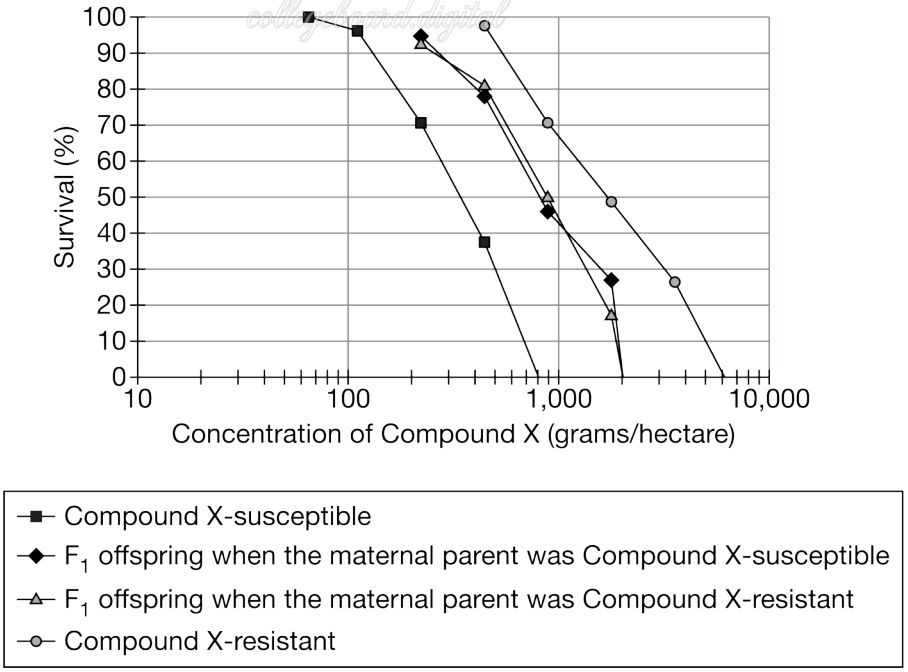


Figure 3. Percent survival of four populations of *L. rigidum* exposed to various concentrations of Compound X shown in a log scale. Data are an average from one experiment with four replicates conducted on seedlings growing in pots.
1 Hectare $= 10{,}000\ \mathrm{m}^2$

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(a) **Explain** how Compound X can affect binding of S3P and PEP to EPSP synthase.

(b) **Identify** one dependent variable in the experiment shown in Figure 2. **Identify** a control group in the experiment shown in Figure 3. Using the data provided in Figure 3, **identify** the concentration of Compound X, in grams/hectare, required to kill 50% of the Compound X-susceptible (S) population. Using the graph in Figure 2, **calculate** the mass, in pg, of EPSP that would be produced by this reaction in one minute at a PEP concentration of 50 µM and a Compound X concentration of 18 µM.

(c) Researchers claim that a single gene is responsible for the Compound X-resistance phenotype. In a follow-up experiment with *L. rigidum*, the researchers crossed an F_1 male with a female from the original resistant (R) population. The resulting offspring were treated with 900 grams/hectare of Compound X. Based on the data presented in Figure 3, **predict** ONE genotype of the resulting offspring and the percent survival of the offspring with this genotype after treatment with Compound X. Researchers claim that if the Compound X-resistance gene were carried on a chloroplast chromosome, the F_1 offspring would not have a phenotype intermediate to that of the two parents. Provide reasoning to **justify** the claim. Researchers discovered a variant of the Compound X-resistance gene in *L. rigidum* in which there was a single base pair insertion upstream of the promoter region of one copy of the gene. **Predict** the most likely effect on the phenotype of the plant.

(d) Researchers have engineered Compound X-resistant corn using prokaryotic genes. Other genetic engineering experiments have placed eukaryotic genes into prokaryotic cells for mass production of their gene products. **Explain** how the eukaryotic DNA would need to be altered in order for prokaryotic cells to express the eukaryotic gene. **Explain** why the *EPSP* synthase gene from a prokaryote can be properly expressed in a corn plant.

5. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Solid tumors are clusters of cancer cells and often contain blood vessels. When molecule B binds to the wild-type Brec protein in the plasma membrane of certain solid tumor cancer cells (Figure 1), the cancer cells express the membrane protein A and sometimes stimulate increased growth of blood vessels into the tumors.

Cells with a particular mutation in the *Brec* gene (Brec-MUT cells) have much increased expression levels of A and stimulate greater growth of blood vessels than do cancer cells with the wild-type Brec (Brec-WT cells); the cells with the mutant Brec can trigger intracellular signaling in the absence of B.

Researchers proposed that the signaling pathway modeled in Figure 1 is triggered by activation of the wild-type Brec and is associated with phosphorylation and activation of kinase D, expression of A, and the ability of the cancer cells to stimulate blood vessel growth.

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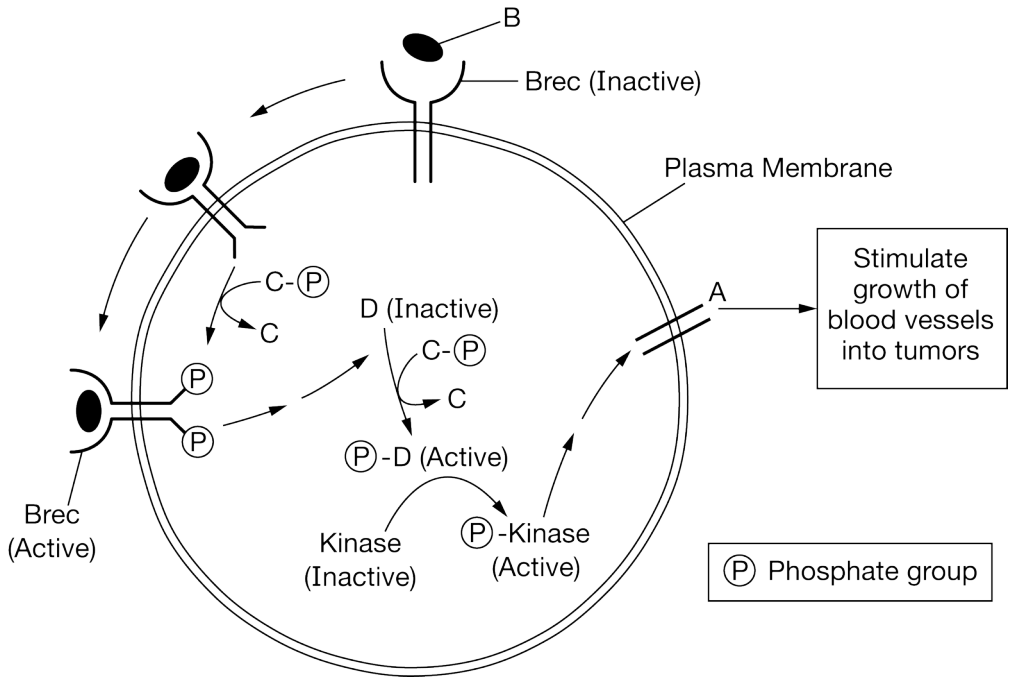


Figure 1. A simplified model of the normal signaling pathway hypothesized to play a role in certain cancer cells expressing A and stimulating blood vessel growth into solid tumors

(a) Based on the signaling model shown in Figure 1, **describe** the role of molecule C-P in the activation of D.

(b) **Explain** how a molecule such as kinase D can amplify an extracellular signal. **Explain** how the mutant Brec protein is able to activate kinase D even in the absence of its ligand.

(c) Based on the proposed signaling pathway, **predict** the relative activity of kinase D in the Brec-WT cells compared with kinase D in the Brec-MUT cells when both cell types are grown in nutrient broth lacking B. **Justify** your prediction.

(d) The researchers claim that treatment with a compound that inhibits dephosphorylation of the intracellular domain of the Brec protein will inhibit expression of A in Brec-WT cells but not in Brec-MUT cells. Based on the information provided, provide reasoning to **refute** their claim.

6. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Glycogen is a complex polymer of glucose molecules involved in the storage of energy in many organisms. Depending on the signals received by a cell, different enzymes are activated, which will lead to either the synthesis or breakdown of glycogen. The product of glycogen breakdown, glucose 1-phosphate, can be modified into glucose 6-phosphate, an intermediate of glycolysis (Figure 1).

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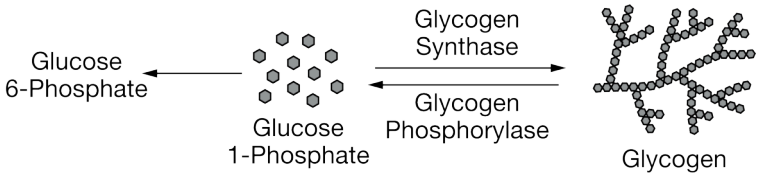


Figure 1. Glycogen synthesis and breakdown

In the liver, glycogen synthase and glycogen phosphorylase are regulated by kinases. These kinases transfer phosphate groups from ATP to amino acids in glycogen synthase and glycogen phosphorylase. Phosphorylase kinase B (PKB) is a kinase that regulates glycogen synthase kinase 3 (GSK3) activity. GSK3 regulates the activity of glycogen synthase. Phosphorylase kinase (PK) regulates the activity of glycogen phosphorylase. The activity of these kinases depends on the activation of signaling pathways in response to the binding of specific cell signals, including insulin and glucagon, to their receptors (Figure 2).

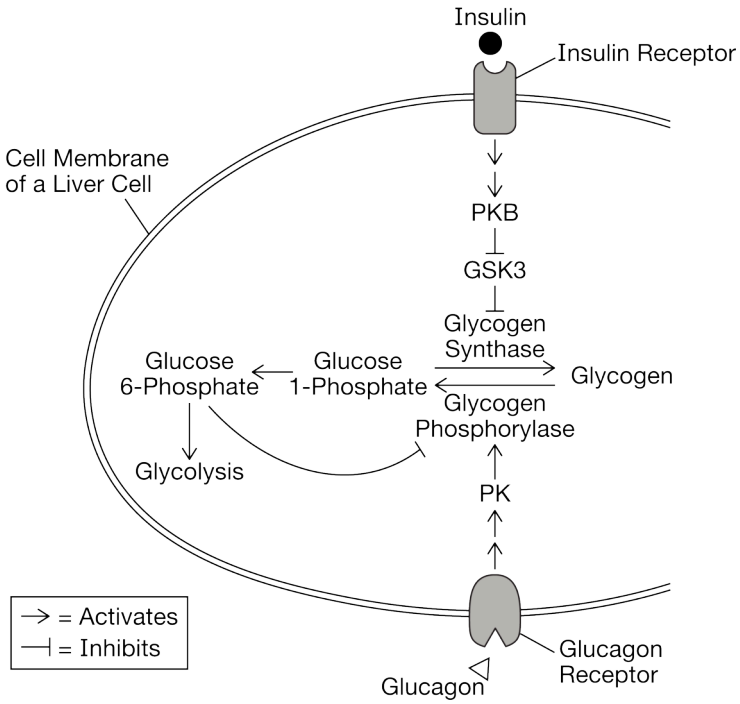


Figure 2. A simplified model of signaling pathways in a liver cell that are activated by insulin and glucagon

(a) A cell where PKB is always active has a mutation that results in the permanent activation of GSK3. Based on the information in Figure 2, **predict** the effect of this mutation on the activity of glycogen synthase. **Justify** your prediction.

(b) **Describe** the process that results in the activation of multiple copies of PKB in response to the binding of a single molecule of insulin to its receptor. **Explain** why insulin can stimulate the activation PKB of but not the activation of PK.

(c) **Explain** why an increase in glycogen phosphorylase activity might result in an increase in O_2 consumption in the cell.

(d) A certain type of tumor results in the overproduction of glucagon. Researchers claim that treatment with insulin can counteract the effects of the excess glucagon on the pathway shown in Figure 2. Provide reasoning to **justify** the researchers' claim.

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7. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

The SIRT3 protein has been reported to increase the growth of certain types of cancer cells. Researchers studying the metabolism and development of stomach cancer cells investigated the potential role of the SIRT3 protein in these cells. In the first investigation, the researchers quantified the relative amount of SIRT3 in normal stomach cells and in each of four different lines of stomach cancer cells (Figure 1). Using data from a group of 100 patients with stomach cancer, the researchers also analyzed the relative amount of SIRT3 protein detected in samples of the cancer cells and in normal cells immediately adjacent to the cancer cells (Figure 2).

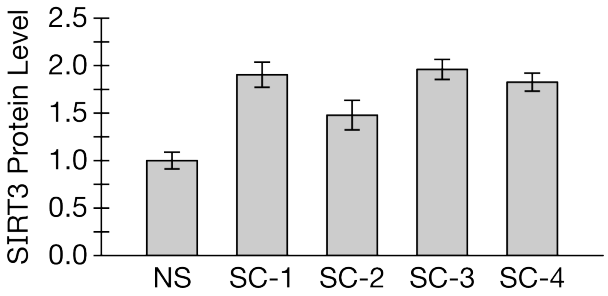


Figure 1. SIRT3 protein levels in normal stomach cells (NS) and stomach cancer cell lines from four different patients (SC-1, SC-2, SC-3, SC-4), shown relative to the level in normal stomach cells (NS). Error bars represent $\pm SE_{\bar{x}}$.

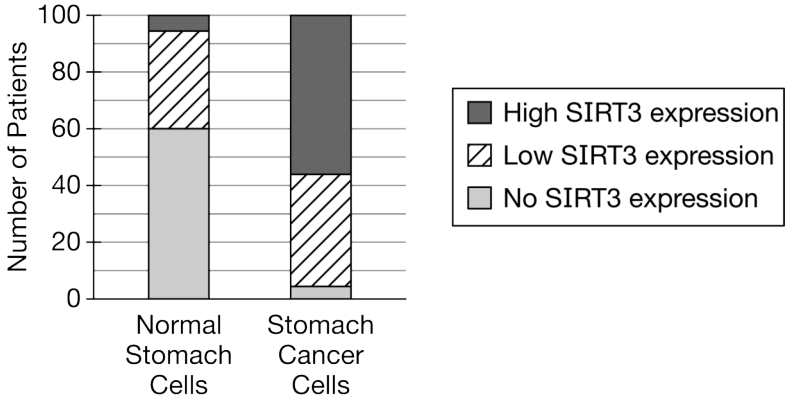
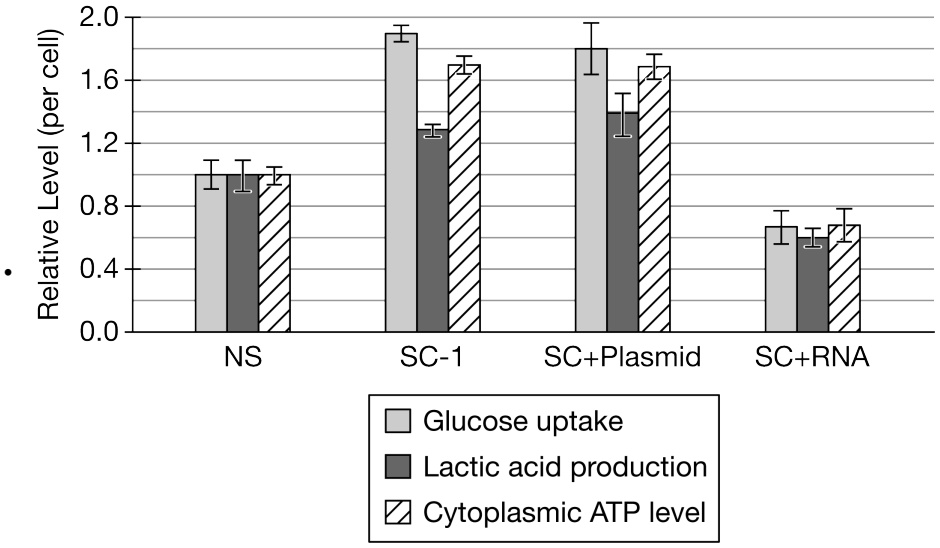


Figure 2. The number of patients, from 100 tested, whose stomach cancer cells or normal stomach cells adjacent to the cancer cells express high, low, or no SIRT3 protein

- In the second investigation, the researchers grew four groups of stomach cells in petri dishes containing nutrient broth:• NS, normal stomach cells• SC-1, a line of stomach cancer cells that the researchers had examined for SIRT3 expression• SC + RNA, the SC-1 line of stomach cancer cells to which a plasmid was added that encodes a small RNA complementary to a portion of the *SIRT3* mRNA. When these small RNAs bind to the complementary portion of *SIRT3* mRNA, the *SIRT3* mRNA is broken down rather than translated.• SC + plasmid, the SC-1 line of stomach cancer cells to which a plasmid was added that does NOT encode the complementary small RNA
- After 24 hours of growth, the researchers measured several indicators of metabolism by an equal number of all four cell groups: glucose uptake, lactic acid production, and the level of cytoplasmic ATP production in the presence of a compound that blocks the electron transport chain (Figure 3). The values were calculated relative to the values obtained for normal stomach cells.

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- Figure 3. Glucose uptake, lactic acid production, and cytoplasmic ATP levels in stomach cells. NS, normal stomach cells; $\mathrm{SC\text{-}1}$, stomach cancer cells; $\mathrm{SC+\ plasmid}$, $\mathrm{SC\text{-}1}$ cells with added plasmid only; $\mathrm{SC\ +\ RNA}$, $\mathrm{SC\text{-}1}$ cells with added plasmid encoding small RNA complementary to a portion of *SIRT3*. Error bars represent $\pm \mathrm{SE}_{\bar{x}}$.

- (a) A stomach cell line growing in nutrient broth has a mutation in the gene encoding a particular type of tRNA. Assuming the mutation is not lethal to the cells, **explain** why the amount of *SIRT3* mRNA in this cell line is identical to the amount produced by a normal stomach (NS) cell line but the amount of $\mathrm{SIRT3}$ protein is less than in the NS cells.

(b) **Identify** a dependent variable in the researchers' experiment shown in Figure 3. **Justify** the analysis of normal stomach cells (NS) in the experiments shown in Figure 3. Based on the data in Figures 1 and 3, **describe** the relationship between $\mathrm{SIRT3}$ expression and glucose uptake. **Calculate** the percent change in lactic acid production by $\mathrm{SC\text{-}1}$ cells compared with $\mathrm{SC+RNA}$ cells.

(c) Researchers analyzing regulation of *SIRT3* gene expression engineer a *SIRT3* gene that is identical to the normal gene except that the normal *SIRT3* gene promoter is replaced by the promoter for a gene encoding a protein typically found only in human nerve cells. The researchers introduce this gene into normal human stomach cells to determine how much *SIRT3* mRNA AND protein will be expressed from the engineered *SIRT3* gene in the stomach cells. **Predict** the results of this experiment. The researchers claim that metabolic activity increases in cancerous cells. Provide reasoning based on the data provided to **justify** the claim. Based on the information provided, **predict** the cause of a decrease in $\mathrm{CO_2}$ production by the three groups of cancer cells ($\mathrm{SC\text{-}1}$, $\mathrm{SC\ +\ plasmid}$, and $\mathrm{SC\ +\ RNA}$) treated with the compound that blocks the electron transport chain compared with the same groups of cells not treated with the compound.

(d) Based on the data provided, **explain** how testing for *SIRT3* gene expression in samples of stomach cells might be used by physicians to predict which individuals have an increased risk of developing stomach cancer. Human stomach cells have plasma membrane receptors for the binding of certain extracellular signaling molecules such as growth factors. **Explain** why the irreversible binding of chemicals that mimic the growth factors will most likely change the normal cells into cancer cells.

8. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Friedreich's ataxia (FA) is an autosomal recessive genetic disorder that causes a lack of muscle control in patients. Heart defects are common in FA patients. FA is associated with a mutated *FXN* gene (*f* allele) that results in decreased levels of the mitochondrial protein frataxin, the *FXN* gene product. Muscle cells in individuals with FA produce decreased amounts of ATP . The normal *FXN* gene contains several copies of a triplet of nucleotides in an intron. Mutant alleles of the gene have

many more copies of the nucleotide triplet, which causes the FXN gene to be abnormally condensed.

To investigate the effects of the FXN mutation, researchers studied heart cells from homozygous dominant individuals (FF) without FA (genotype A), heart cells from patients with FA (genotype B), and heart cells from individuals heterozygous for the FXN gene (genotype C). The researchers isolated two different proteins from an equal number of each of the three cell genotypes. For each cell genotype, they compared the amount of frataxin protein with the amount of glyceraldehyde 3-phosphate dehydrogenase (GAPDH), an enzyme that functions in glycolysis (Figure 1). They then quantified the relative amount of frataxin protein in each cell genotype (Figure 2).

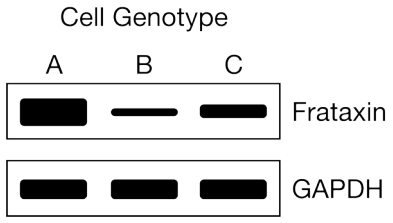


Figure 1. Amounts of frataxin and GAPDH proteins in the three different cell genotypes A, B, and C, as revealed by band width following protein gel electrophoresis

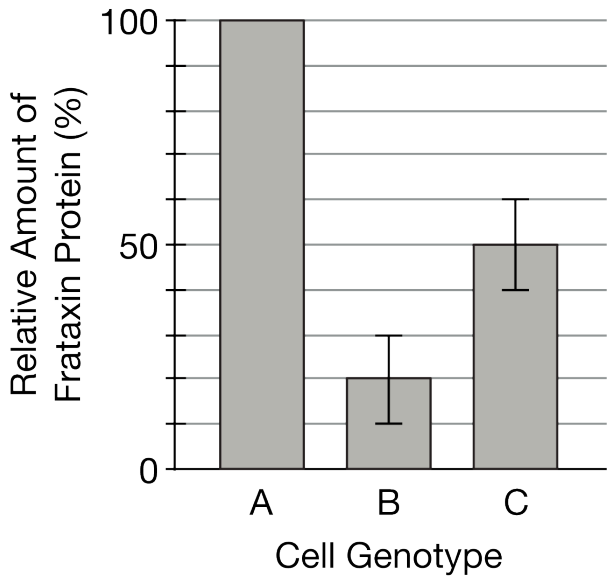


Figure 2. Quantified relative amount of frataxin protein in cell genotypes A, B, and C. Data represent the average from six trials, as shown in Figure 1. Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

To investigate a potential role for frataxin in cellular respiration, another group of researchers introduced the human FXN gene into a mouse cell line and overexpressed the frataxin protein. Identical control cells were similarly treated but did not receive the human FXN gene. The researchers measured the activity of a Krebs cycle enzyme in an equal number of control cells and cells overexpressing the human FXN gene (Figure 3). In a related experiment, an equal number of both cell types were grown in medium with $^{14}\mathrm{C}$-labelled glucose, and the researchers measured the amount of $^{14}\mathrm{C}$-labelled CO_2 produced by the cells (Figure 3). They also determined the concentration of lactic acid in both cell types (Figure 3).

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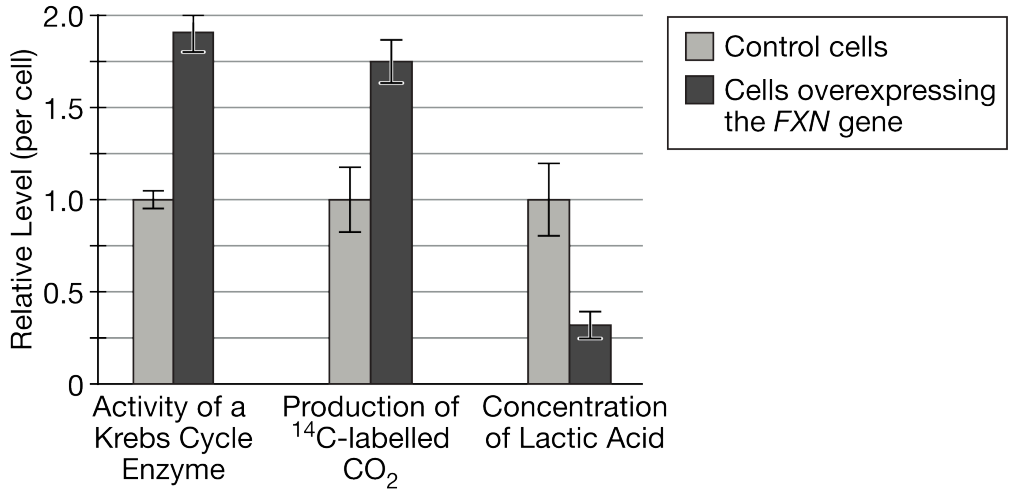


Figure 3. The activity of a Krebs cycle enzyme, the production of $^{14}\mathrm{C}$-labelled CO_2 after growth in a medium with $^{14}\mathrm{C}$-labelled glucose, and the concentration of lactic acid in control cells and in cells overexpressing the FXN gene. Values are shown relative to the values in the control cells. Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

(a) **Describe** the possible genotypes of the heart cells of an individual if meiosis in this individual results in two sperm cells carrying the f allele and two sperm cells carrying the F allele.

(b) **Identify** a dependent variable for the experiment with data shown in Figure 2. **Justify** the measurement of $^{14}\mathrm{C}$-labelled CO_2 in the experiment with data shown in Figure 3. **Describe** the difference between the mouse cells overexpressing the FXN gene and the control mouse cells with respect to the $^{14}\mathrm{C}$-labelled CO_2 production shown in Figure 3. Based on the data in Figure 3, **calculate** how many times more lactic acid is produced on average in the control cells as compared with cells overexpressing FXN.

(c) Researchers produced mice that contained three copies of the FXN gene in each cell, producing genotype fff, rather than the typical two copies. Based on the data provided, **predict** the amount of frataxin protein most likely to be produced by the mice compared with the amount of frataxin in cells with genotypes A, B, and C. **Justify** your prediction.

(d) **Explain** why the phenotypes associated with the human FXN gene can be studied in a mouse cell. Unlike skeletal muscle cells, heart muscle cells cannot rely on fermentation for long. **Explain** the inability of heart muscle cells to function using only fermentation for metabolism. ATP acts as a negative feedback regulator of phosphofructokinase, one of the enzymes in glycolysis. **Predict** the eventual effect of ATP on pyruvate production in heart cells that express excess amounts of the normal frataxin.

9. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

The $\mathrm{SIRT3}$ protein has been reported to increase the growth of certain types of cancer cells. Researchers studying the metabolism and development of stomach cancer cells investigated the potential role of the $\mathrm{SIRT3}$ protein in these cells. In the first investigation, the researchers quantified the relative amount of $\mathrm{SIRT3}$ in normal stomach cells and in each of four different lines of stomach cancer cells (Figure 1). Using data from a group of 100 patients with stomach cancer, the researchers also analyzed the relative amount of $\mathrm{SIRT3}$ protein detected in samples of the cancer cells and in normal cells immediately adjacent to the cancer cells (Figure 2).

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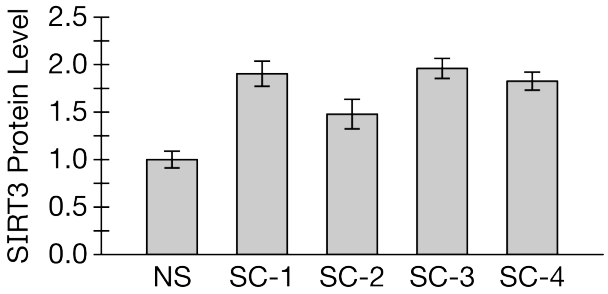


Figure 1. SIRT3 protein levels in normal stomach cells (NS) and stomach cancer cell lines from four different patients (SC-1, SC-2, SC-3, SC-4), shown relative to the level in normal stomach cells (NS). Error bars represent $\pm SE_{\bar{x}}$.

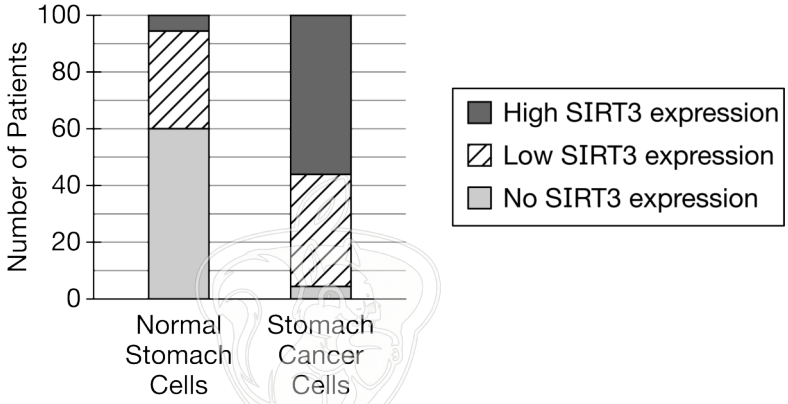
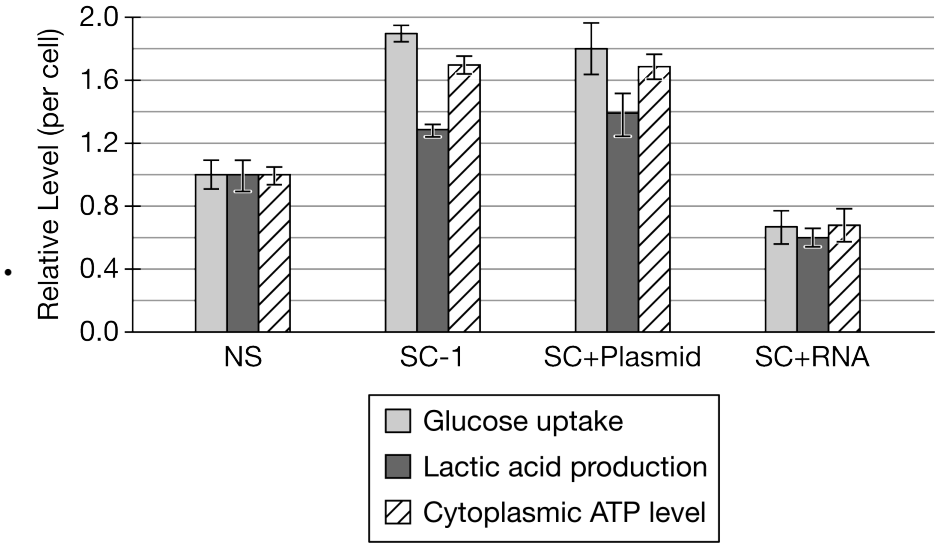


Figure 2. The number of patients, from 100 tested, whose stomach cancer cells or normal stomach cells adjacent to the cancer cells express high, low, or no SIRT3 protein

- In the second investigation, the researchers grew four groups of stomach cells in petri dishes containing nutrient broth:• NS, normal stomach cells• SC-1, a line of stomach cancer cells that the researchers had examined for SIRT3 expression• SC + RNA, the SC-1 line of stomach cancer cells to which a plasmid was added that encodes a small RNA complementary to a portion of the *SIRT3* mRNA. When these small RNAs bind to the complementary portion of *SIRT3* mRNA, the *SIRT3* mRNA is broken down rather than translated.• SC + plasmid, the SC-1 line of stomach cancer cells to which a plasmid was added that does NOT encode the complementary small RNA
- After 24 hours of growth, the researchers measured several indicators of metabolism by an equal number of all four cell groups: glucose uptake, lactic acid production, and the level of cytoplasmic ATP production in the presence of a compound that blocks the electron transport chain (Figure 3). The values were calculated relative to the values obtained for normal stomach cells.

FRQ



- Figure 3. Glucose uptake, lactic acid production, and cytoplasmic ATP levels in stomach cells. NS, normal stomach cells; $\mathrm{SC\text{-}1}$, stomach cancer cells; $\mathrm{SC+\ plasmid}$, $\mathrm{SC\text{-}1}$ cells with added plasmid only; $\mathrm{SC\ +\ RNA}$, $\mathrm{SC\text{-}1}$ cells with added plasmid encoding small RNA complementary to a portion of $\mathit{SIRT3}$. Error bars represent $\pm \mathrm{SE}_{\bar{x}}$.

- (a) **Explain** why an mRNA transcribed from the template strand of the $\mathit{SIRT3}$ gene in normal stomach (NS) cells is translated to produce a $\mathrm{SIRT3}$ protein of correct sequence, but if an mRNA were transcribed from the nontemplate strand of the $\mathit{SIRT3}$ gene in these cells, it would not produce a $\mathrm{SIRT3}$ protein of correct sequence.

(b) **Identify** a control group for the analysis shown in Figure 3. **Justify** analyzing $\mathrm{SIRT3}$ protein level in four different cancer cell lines, as shown in Figure 1. Based on Figures 1 and 3, **describe** the relationship between $\mathrm{SIRT3}$ expression and cytoplasmic ATP levels. **Calculate** the percent change in cytoplasmic ATP levels by $\mathrm{SC\ +\ RNA}$ cells compared with $\mathrm{SC\ +\ plasmid}$ cells.

(c) Researchers analyzing the $\mathit{SIRT3}$ gene sequence engineer a $\mathit{SIRT3}$ gene that is identical to the normal gene except that a single nucleotide has been deleted from the $5'$ end of the second exon of the gene; the deletion does not affect splicing. The researchers introduce this gene into normal stomach cells to determine how the primary AND tertiary structures of the protein expressed from the engineered $\mathit{SIRT3}$ gene will compare with the protein expressed from the normal $\mathit{SIRT3}$ gene in these cells. **Predict** the results of this experiment. The researchers claim that the data in Figure 3 provide evidence that the stomach cancer cells perform fermentation to a greater extent than do normal stomach cells. Provide reasoning to **justify** the claim. Based on the information provided, **predict** the effect on the ratio of $\mathrm{NADH/NAD^+}$ in the cells treated with the compound that blocks the electron transport chain compared with similar cells not treated with the compound.

(d) Based on the data provided, **explain** how cellular processes must be altered in stomach cancer cells compared with normal stomach cells to result in the different levels of $\mathrm{SIRT3}$ expression observed. A mutation to the gene encoding a cyclin-dependent kinase (CDK) in human stomach cells results in a CDK that is continually active in the cells. **Explain** why the continually active CDK will most likely change the normal cells into cancer cells.

10. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Hemoglobin is an iron-containing protein in red blood cells that transports oxygen from the lungs to cells throughout the body. Hemoglobin is a protein complex that is composed of four polypeptide subunits. Typically, two of the polypeptide subunits are encoded by the $\mathit{HBA1}$ gene and are called alpha-globin subunits, while the other two polypeptide subunits are encoded by the HBB gene and are called beta-globin subunits. Each globin subunit contains a single iron atom capable of binding oxygen.

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Prior to birth and during early infancy, expression of the HBB gene is low and, instead, cells produce a third polypeptide called gamma-globin. During this time, the hemoglobin complex consists of two alpha-globin subunits and two gamma-globin subunits and is referred to as fetal hemoglobin. Fetal hemoglobin is able to bind oxygen more tightly than does the typical adult hemoglobin, which enables the fetus (unborn offspring) to remove oxygen from the hemoglobin in the mother's blood. A model of the expression levels of all three types of globin polypeptides during the weeks leading up to and following birth is shown in Figure 1.

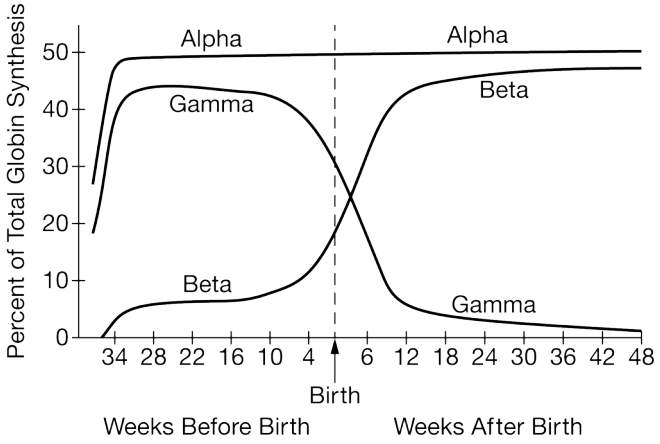


Figure 1. Model of globin expression in fetal and infant tissues

Mutations in the genes that encode globin subunits may affect the function of hemoglobin. Beta-thalassemia is an inherited disorder that results from a variety of mutations to the HBB gene that encodes the beta-globin subunit. The incidence of beta-thalassemia in one particular family is shown in Figure 2.

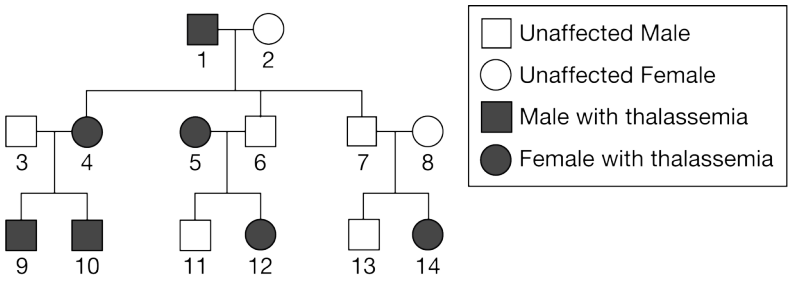


Figure 2. Incidence of beta-thalassemia in three generations of a family

The inability to produce functional beta-globin results in an imbalance of alpha- and beta-globin subunits. The accumulation of free alpha-globin subunits causes red blood cells to lyse (burst) or undergo apoptosis (Figure 3). The extra alpha-globin subunits are unable to correctly fold into a hemoglobin protein complex and instead form structures called alpha-hemichromes. The alpha-hemichromes are unable to bind oxygen molecules and become insoluble in the cytoplasm of the red blood cells. The alpha-hemichromes are then deposited in solid form on the surface of the red blood cells, causing the red blood cells to lyse.

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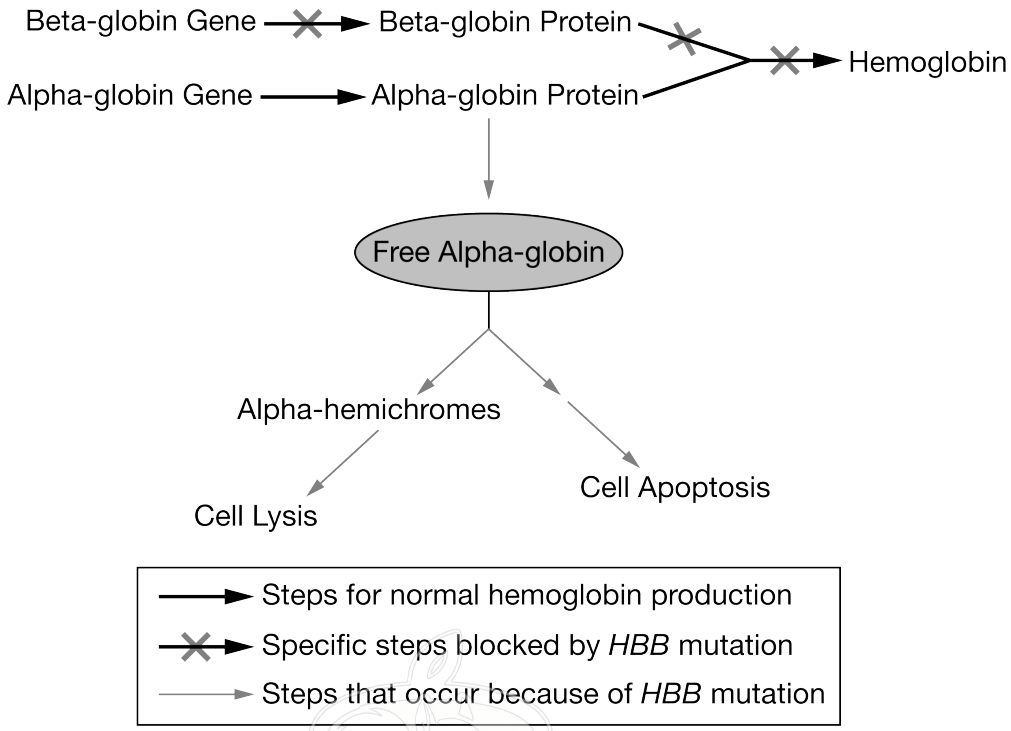


Figure 3. Model of globin synthesis and hemoglobin assembly and the effects of beta-thalassemia. Black arrows with gray Xs indicate steps in the normal hemoglobin assembly that are blocked by the HBB mutation. Gray arrows indicate steps that occur because of the HBB mutation.

(a) Based on the pedigree shown in Figure 2, **describe** the HBB alleles inherited by individual 6, including the parent from which each allele came.

(b) **Explain** why the symptoms associated with beta-thalassemia are often undetectable at time of birth. A study of the gene encoding the gamma-globin subunit showed that, shortly after birth, the gene becomes methylated (CH_3 groups are added to the DNA). **Explain** how DNA methylation could cause the change in the amount of gamma-globin protein that is represented in Figure 1.

(c) **Predict** the effect of a treatment that reduces the expression of $HBA1$ on the symptoms of individuals with beta-thalassemia. **Justify** your prediction.

(d) Researchers claim that the ability of fetal hemoglobin to bind oxygen more tightly than the hemoglobin of the mother increases the fitness of the fetus. Provide reasoning to **support** the claim.

11. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Solid tumors are clusters of cancer cells and often contain blood vessels. When molecule B binds to the wild-type Brec protein in the plasma membrane of certain solid tumor cancer cells (Figure 1), the cancer cells express the membrane protein A and sometimes stimulate increased growth of blood vessels into the tumors.

Cells with a particular mutation in the $Brec$ gene (Brec-MUT cells) have much increased expression levels of A and stimulate greater growth of blood vessels than do cancer cells with the wild-type Brec (Brec-WT cells); the cells with the mutant Brec can trigger intracellular signaling in the absence of B.

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Researchers proposed that the signaling pathway modeled in Figure 1 is triggered by activation of the wild-type Brec and is associated with phosphorylation and activation of kinase D, expression of A, and the ability of the cancer cells to stimulate blood vessel growth.

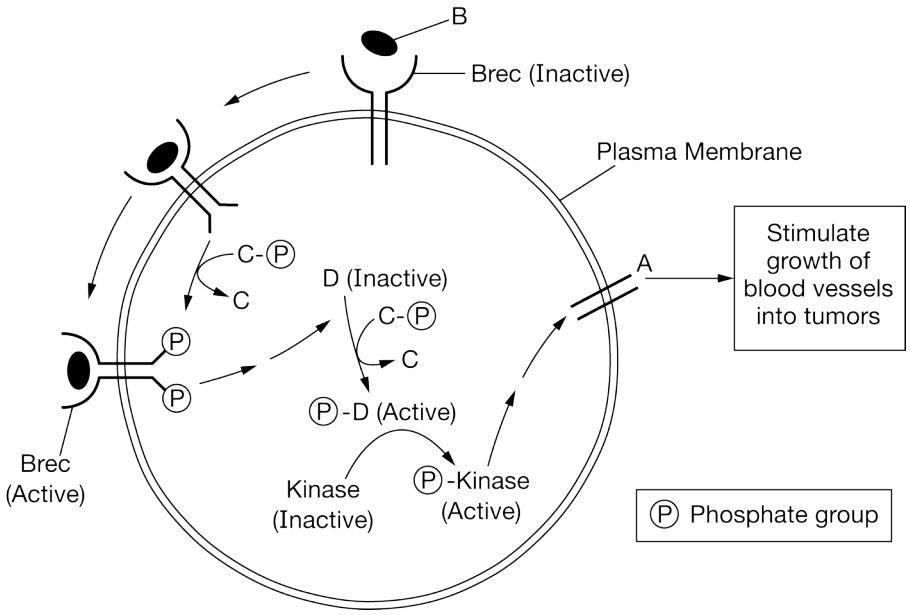


Figure 1. A simplified model of the normal signaling pathway hypothesized to play a role in certain cancer cells expressing A and stimulating blood vessel growth into solid tumors

(a) Based on the signaling model shown in Figure 1, **describe** the role of kinase D.

(b) A certain group of Brec-MUT cells expresses low levels of protein A, unlike the high expression levels of most Brec-MUT cells. Researchers determine that the cells have an excess of ATP and that there are no mutations in the gene encoding protein A. **Describe** a change in the Brec signaling pathway that would lead to the decreased expression of protein A even in the presence of Brec-MUT. **Explain** why B is able to bind to Brec but not to A.

(c) Based on the proposed signaling pathway, **predict** the relative amount of phosphorylated to unphosphorylated mutant Brec in the Brec-MUT cells when the cells are grown in nutrient broth lacking B. **Justify** your prediction.

(d) The researchers claim that treatment with a compound that inhibits phosphorylation of kinase D will affect expression of A in the Brec-WT cells but not in the Brec-MUT cells. Based on the information provided, provide reasoning to **refute** their claim.

12. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Human hair grows in a cycle that includes a growth phase and a rest phase. A small fraction of the hairs on the scalp are in the rest phase, while the majority of the hairs on the scalp are in the growth phase, which allows the hair on the scalp to grow long. Conversely, the majority of hairs in areas such as the eyelashes are in the rest phase, which keeps those hairs short.

Familial trichomegaly (FT) is a genetic condition that results in excessively long eyelashes in affected individuals. In an investigation into the causes of FT, a team of researchers identified mutations in the gene that codes for fibroblast growth factor-5 (FGF5) that are associated with FT. FGF5 is a protein that is secreted by certain cells and functions as a signaling molecule that influences the ratio of growth phase to rest phase in hair. The structure of the *FGF5* gene includes 3 exons and 2

introns. Alternate splicing of the pre-mRNA transcribed from the gene creates a long version of the mRNA, *FGF5-L*, and a short version of the mRNA, *FGF5-S* (Figure 1).

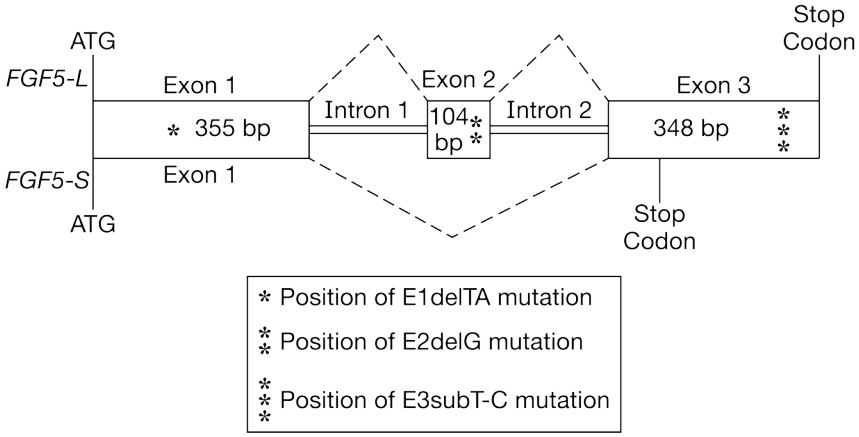
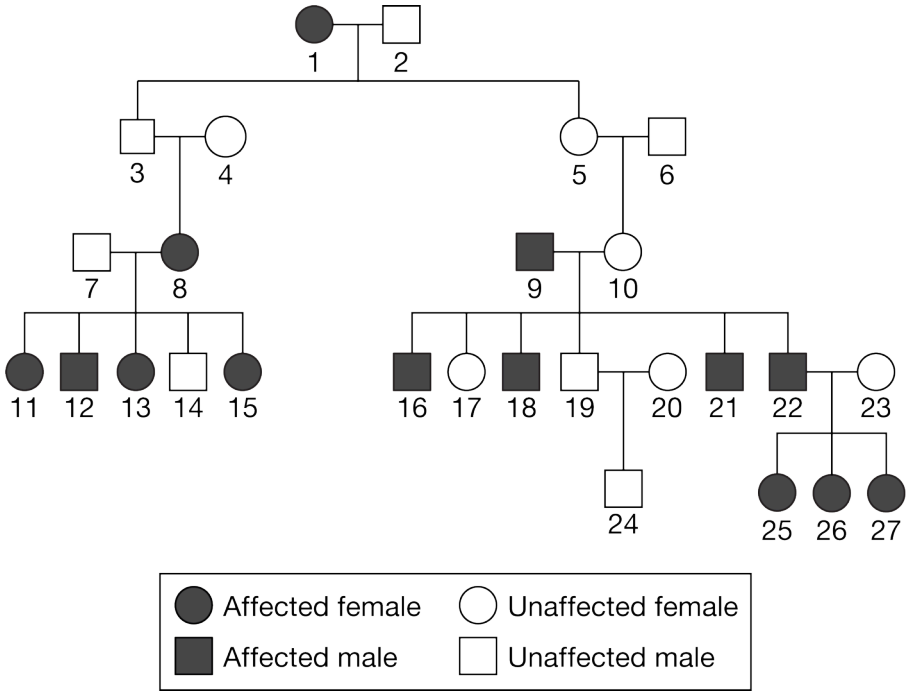


Figure 1. A map of the splicing patterns that lead to the production of both the FGF5-L protein and the FGF5-S protein. Labels below the diagram refer to gene components for *FGF5-S*, and labels above the diagram refer to gene components for *FGF5-L*. Dotted lines indicate the portion(s) of the sequence that will be removed from the transcribed pre-mRNA to form either *FGF5-L* or *FGF5-S* mRNA .

- Researchers identified three different mutations in the *FGF5* gene that have been associated with FT. These mutations, the positions of which are shown in Figure 1, include: • E1delTA: a mutation in which a thymine and an adenine are deleted from Exon 1 of the *FGF5* gene• E2delG: a mutation in which a guanine is deleted from Exon 2 of the *FGF5* gene• E3subT-C: a mutation in which a thymine is substituted for a cytosine in Exon 3 of the *FGF5* gene

The researchers collected data on a family with a history of FT and created a pedigree representing the instances of FT in the family as shown in Figure 2.



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Figure 2. A pedigree representing five generations of a family with a history of FT

The researchers analyzed the length of hairs from individuals with FT and from unaffected individuals and plotted the length of each hair as shown in Figure 3A. They also determined the ratio of hairs in growth phase to hairs in rest phase as shown in Figure 3B.

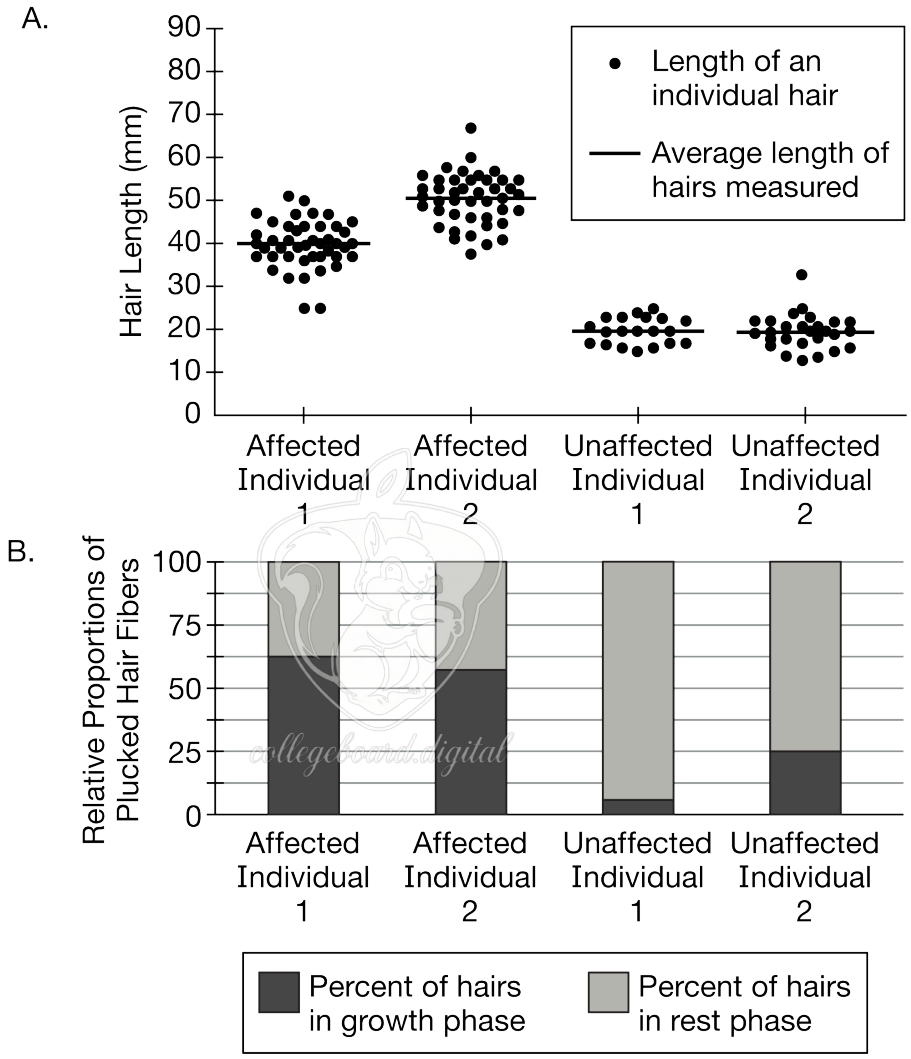


Figure 3A. Length of hairs taken from individuals with and without FT
Figure 3B. Proportion of hairs examined that were in growth phase compared with the proportion of examined hairs in rest phase

In a final experiment, the researchers grew human hair in a petri dish with and without adding FGF5 to the dish. The growth of the hair over an 8-day period was measured. The data from this experiment are shown in Figure 4.

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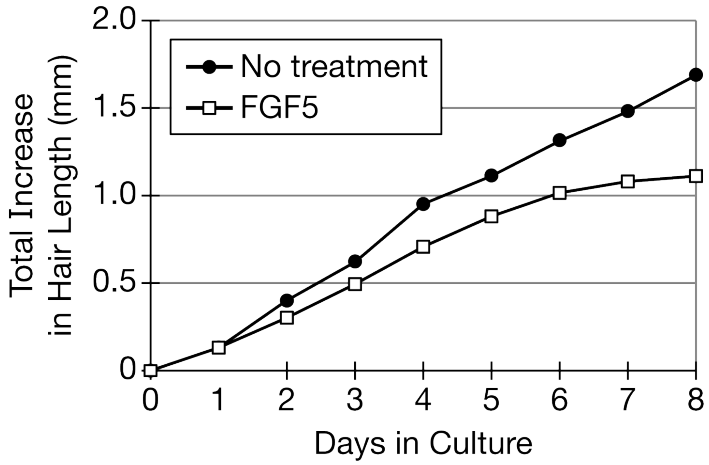


Figure 4. Growth of human hair in culture with and without FGF5

(a) **Describe** what happens to an *FGF5* mRNA to which a poly(A) tail has not been added.

(b) **Identify** a dependent variable in the experiment represented in Figure 4. **Justify** using the culture without FGF5 as the control in the experiment represented in Figure 4. Based on an analysis of the pedigree provided in Figure 2, **describe** the most likely mode of inheritance of FT in this family. Based on Figure 2, if individuals 3 and 4 had another child, **calculate** the probability that the child would have been an affected female.

(c) **Predict** the effect of the E3subT-C mutation on the function of FGF5-S. **Justify** your prediction. Researchers identify a new mutation that results in an FGF5 receptor that can bind the FGF5 protein but cannot stimulate production of second messengers in the cell. Based on the data provided in Figure 4, **predict** the effect of the new mutation on hair growth, regardless of hair type.

(d) Researchers introduced a single copy of the normal mouse *FGF5* allele into human cells homozygous for the *FGF5* mutation and found that, while the human cells could express the mouse *FGF5* gene, the mouse FGF5 protein was not able to restore normal FGF5 signaling in the human cells. **Explain** how the mouse FGF5 protein could be expressed but not activate the FGF5 pathway in the human cells. Bacterial cells can be used to produce large quantities of eukaryotic proteins. **Explain** why a functional FGF5 protein is not produced when bacteria are transformed with the full-length eukaryotic *FGF5* gene, even when the eukaryotic promoter is replaced with a bacterial promoter.

13. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Crossing-over between homologous chromosomes during meiosis is necessary for the proper segregation of the chromosomes. In the fly species *Drosophila melanogaster*, two conserved genes, *mei-9* and *pch2*, are thought to play a role in crossing-over. The protein encoded by *mei-9* is part of a large protein complex required for the pairing of homologous chromosomes, while the protein encoded by *pch2* is associated with a delay in prophase I that is believed to serve as a cell cycle checkpoint.

To investigate the role of *mei-9* and *pch2* in crossing-over, researchers determined the frequency of recombination for each of four regions of chromosome 2 in wild-type flies (WT), flies expressing a mutant form of *mei-9* (mei-9-mut flies), and flies expressing mutant forms of both *mei-9* and *pch2* (mei-9-mut/pch2-mut flies). The frequency of recombination of each region in both types of mutant flies is shown in Figure 1 as a percentage of the frequency of recombination observed in WT flies.

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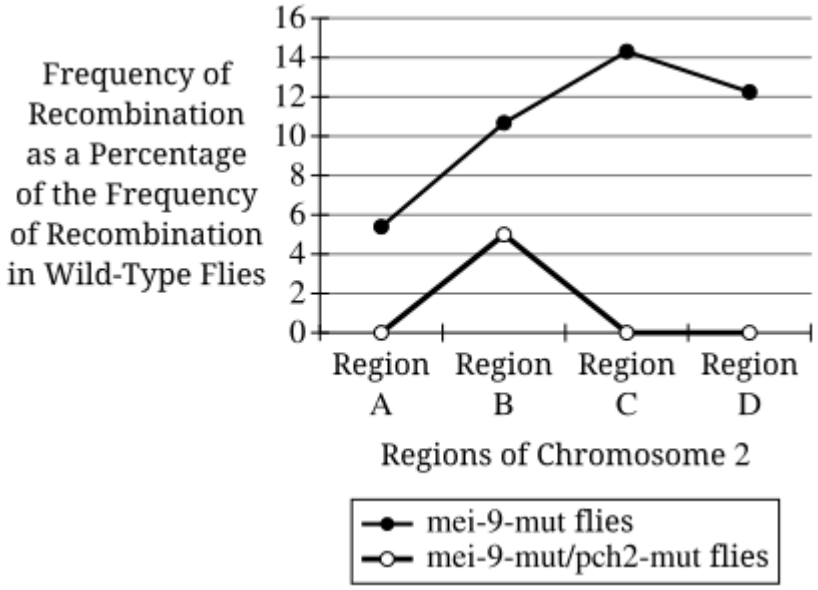


Figure 1. Frequency of recombination of four regions of chromosome 2 in mei-9-mut flies and mei-9-mut/pch2-mut flies as a percentage of the frequency of recombination in the same region in wild-type flies

(a) Based on the data in Figure 1, **identify** the region of chromosome 2 with the greatest frequency of recombination in mei-9-mut/pch2-mut flies.

(b) Based on the data in Figure 1, **describe** the frequency of recombination in mei-9-mut flies compared with the frequency of recombination in WT flies.

(c) Scientists hypothesize that the function of the protein encoded by wild-type *pch2* allows some of the defects caused by the *mei-9* mutation to be overcome, increasing the likelihood that crossing-over will occur. Use the data in Figure 1 to **evaluate** their hypothesis.

(d) Based on the information provided, **explain** why the fertility of mei-9-mut flies might be reduced compared with that of WT flies.

14. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Tuberculosis is an infectious disease caused by the bacterium *Mycobacterium tuberculosis.* Many of the mycobacteria are resistant to the drugs typically used for treatment. As a part of the process of developing new drugs, scientists identified two mycobacterial proteins, L10 and L12, that are part of the ribosome and that must bind to each other to allow ribosomal function. The scientists aimed to identify a compound that blocks the interaction between the two proteins, thereby preventing the survival and reproduction of *Mycobacterium tuberculosis*.

As a first step, the scientists performed an *in vitro* (in a test tube) experiment to test the ability of different compounds to block the binding between the L10 and L12 proteins. The scientists measured the amount of binding in the presence of increasing concentrations of three compounds: I, II, and III. Results are shown in Figure 1.

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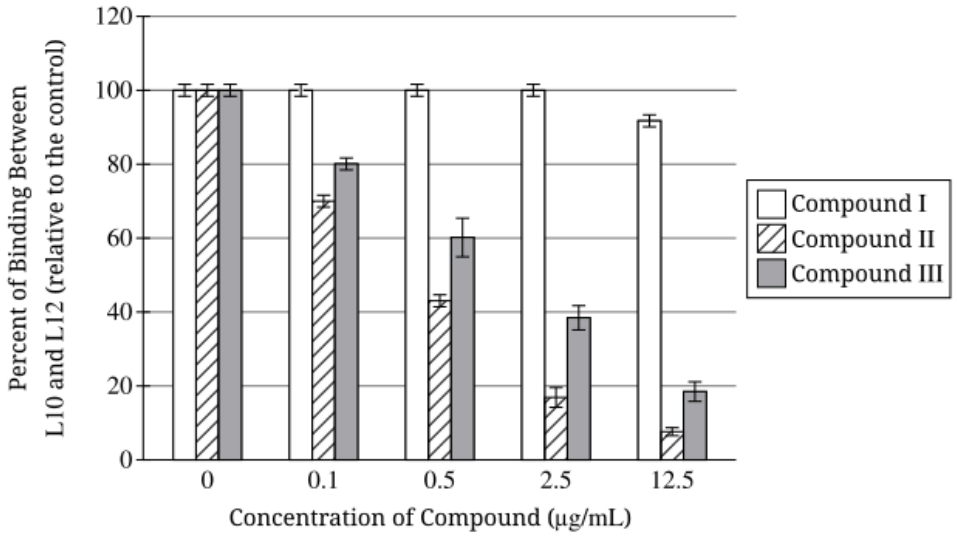


Figure 1. Average percent of binding between $\mathrm{L10}$ and $\mathrm{L12}$ in the presence of increasing concentrations of compounds I, II, and III, relative to the amount of binding in the absence of any compound ($0\ \mu\mathrm{g/mL}$, the control). Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

In a second experiment, the scientists performed an *in vitro* experiment to test the amount of inhibition of translation in the presence of compounds I, II, and III. They added the three compounds to test tubes containing molecules needed for translation, including a specific mRNA, and ribosomes from one of three organisms: *Mycobacterium tuberculosis*, the bacterium *Escherichia coli*, or the eukaryote *Oryctolagus cuniculus* (rabbit). They measured the concentration of each compound required to reduce translation by 50% (Table 1).

TABLE 1. CONCENTRATION ($\mu\mathrm{g/mL}$) REQUIRED TO REDUCE TRANSLATION BY 50% FOR COMPOUNDS I, II, AND III

Compound	*M. tuberculosis* ribosomes	*E. coli* ribosomes	*O. cuniculus* ribosomes
I	162.7	143.1	30.2
II	16.9	117.7	118.1
III	25.9	34.2	60.1

(a) Based on the data in Figure 1, **identify** the concentration in $\mu\mathrm{g/mL}$ of compound III that reduces the amount of binding between $\mathrm{L10}$ and $\mathrm{L12}$ to 60% of the control.

(b) Based on the data in Figure 1, **describe** the specific decrease in $\mathrm{L10}$ and $\mathrm{L12}$ binding in relation to each specific increase in compound III concentration from $0.1\,\mu\mathrm{g/mL}$ to $12.5\,\mu\mathrm{g/mL}$.

(c) Based on the data in Table 1, the scientists predict that compound II is the best of the three compounds to treat tuberculosis infections. **Evaluate** the data to support or refute their prediction.

(d) One of the compounds becomes routinely used as a drug to treat tuberculosis. Based on the experiments, **explain** how a mutation in *Mycobacterium tuberculosis* would most likely cause resistance to the drug.

15. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written

out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Several studies indicate that a particular forest biome in the Northern Hemisphere will become significantly warmer over the next century. Researchers studying a population of trees in the forest hypothesize that warming will negatively impact the evolutionary fitness of many of the trees. The researchers also hypothesize that warming will shift the range of the population further north to cooler habitats.

(a) **Describe** the way the evolutionary fitness of an individual tree is measured.

(b) The range shift of the tree population may result in the fragmentation of the population into several smaller, isolated populations. **Explain** how the fragmentation may result in speciation.

(c) Recent research suggests that the range in which trees can grow is due in large part to their symbiotic relationships with soil microbes that are sensitive to temperature. **Predict** the effects on a population of microbes if the average temperature in an area increases above the temperature at which most of the microbes are able to successfully reproduce.

(d) **Justify** your prediction in part (c) using evidence from biological principles.

16. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

During the Late Devonian Period, there was a massive loss of marine biodiversity. An analysis of fossil marine invertebrates that existed during the Late Devonian has revealed that the rate of extinction was relatively constant, as compared with the rates of extinction before and after the Late Devonian. The fossil evidence also suggests that frequent increases in sea level during the Late Devonian effectively removed geographical barriers and allowed generalist species (species that can inhabit a wide range of habitats) to migrate into new areas.

(a) **Describe** how the geographical separation of two populations can lead to speciation.

(b) **Explain** how a decrease in biodiversity during the Late Devonian could occur even though the extinction rate was relatively constant.

(c) Scientists claim that the modern rate of biodiversity loss may be even greater than that observed during the Late Devonian. In recent decades, the number of invasive species has increased. **Predict** the most likely effect of the current increase in invasive species on speciation rates of native species.

(d) Provide reasoning to **justify** your prediction in part (c).

17. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

A population of invasive feral hogs in a wilderness area is approaching carrying capacity. The feral hogs in each age-group of this population are counted once a year. Figure 1 shows the age pyramid of this population in one particular year. Feral hogs reproduce between 1 and 5 years of age. Mothers care for their young until the young are three months old.

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Figure 1. Age pyramid of a feral hog population in a wilderness area

Ten years later a researcher again counts the number of individual feral hogs in each age-group in the same population and produces the age pyramid shown in Figure 2.

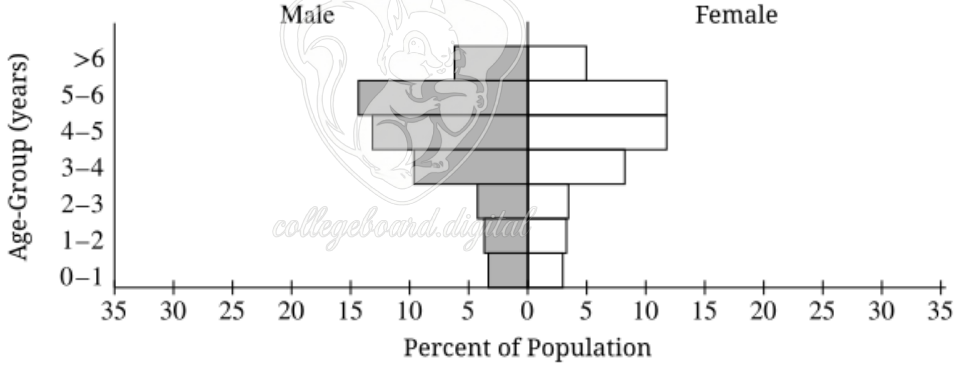


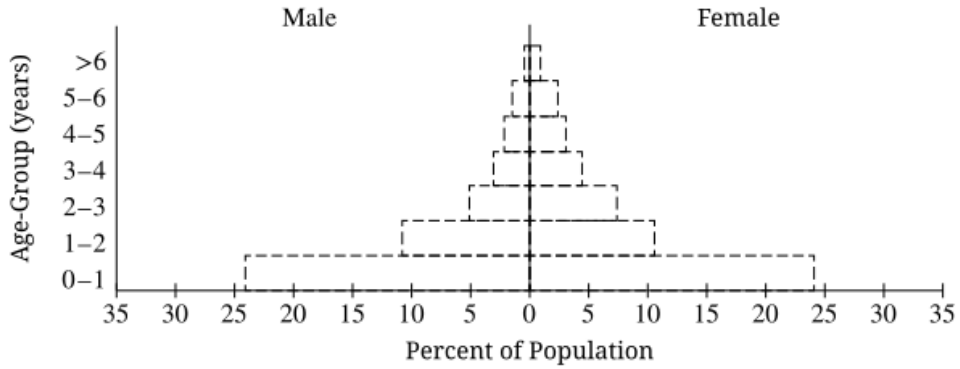
Figure 2. Age pyramid of the feral hog population ten years later

(a) **Describe** a characteristic of an ecosystem that allows an introduced species to become an invasive species.

(b) A population age structure, as shown in Figure 1, is the result of multiple factors affecting birth and death rates of animals of various ages. **Explain** how ecological factors lead to the difference between the numbers of hogs in the $0-1$ year age-group and the number of hogs in the $1-2$ year age-group shown in Figure 1.

(c) An outbreak of a viral infection causes approximately half of the male hogs in each age-group to die. Using the template in the space provided for your response, **draw** new bars for each age-group to represent the approximate distribution of the entire hog population immediately after the death of the males. The dashed lines of the template represent the original age pyramid (Figure 1) before the viral outbreak.

FRQ



(d) Using biological principles, **explain** what is likely to have caused the shift from the age structure in Figure 1 to the age structure shown in Figure 2.

18. Read each question carefully. Write your response in the space provided for each part of each question. Answers must be written out in paragraph form. Outlines, bulleted lists, or diagrams alone are not acceptable and will not be scored.

Vernal ponds, ponds that fill with water each spring and then evaporate in late summer, pose particular challenges to the organisms that live in them. Copepods, a type of crustacean, produce eggs that survive in dry sediment (the soil and other solid material at the bottom of a pond) for long periods of time. Once the water returns, the copepod larvae emerge from the eggs and begin to eat, consuming primarily algae. However, algae often emerge from the sediment later than the copepod larvae do, and the algal populations grow slowly in cool spring temperatures.

To investigate whether the copepods that live in vernal ponds have other adaptations to allow them to supplement their diet of algae, scientists examined related species that they categorized as large-bodied or small-bodied copepods. After examining the mouthparts of each species, the scientists determined a mouthpart carnivory score for each. A high score indicates a tendency toward a more carnivorous diet. The scientists also recorded the gut contents of each species in both early spring and summer. They hypothesized that large-bodied copepods have mouth adaptations that enable them to eat small animals that emerge early from sediments, while the small-bodied copepods do not have such adaptations.

(a) **Describe** the effect of periodic changes in resource availability on population density.

(b) **Justify** the researchers' performing their measurements and analyzing gut contents in both early spring and summer.

(c) **State** a null hypothesis for this investigation.

(d) Climate change is causing water in vernal ponds to evaporate earlier in the summer. Scientists claim that under these conditions, copepod species with a lower mouthpart carnivory score are more likely to go extinct in these vernal ponds than are species with a greater mouthpart carnivory score. Provide reasoning to **support** the scientists' claim.